Explainable AI-Driven Financial Transaction Fraud Detection using Machine Learning and Deep Neural Networks

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A B S T R A C T

Artificial Intelligence (AI) is revolutionizing and transforming data-driven businesses like financial services, e-commerce, and banking. AI technology and the digitization of banking are rapidly increasing the volume of financial transaction frauds, causing billions in revenue losses and unrecoverable funds. The primary business challenge faced by the finance and banking industry is understanding how their AI systems make predictions and explaining why a transaction is flagged as fraud. A solution to the "black box" challenge is known as Explainable Artificial Intelligence (XAI). XAI is a human-centric AI that accelerates business value through a greater understanding of predictions and makes complex black box decisions more transparent, trustable, explainable, and understandable to human users for better business decision-making without compromising prediction accuracy. This research involves the development of a novel classification model to classify whether a financial transaction is fraudulent or legitimate using five Machine learning algorithms, including Decision Tree, Logistic Regression, Random Forest, Light Gradient Boosting Machine (Light GBM), Extreme Gradient Boosting (XGBoost) and two Deep Neural Networks Algorithms like Artificial Neural Networks (ANN) and Convolutional Neural Networks (CNN). To tackle challenges like data quality issues and high-class imbalance in the dataset, several experimentations like Memory Optimisation, Hyperparameter Tuning, Principal Component Analysis (PCA), and Random Oversampling methods have been implemented. Based on the literature, this is a novel instance of research investigating and implementing five XAI methods on the IEEE-CIS fraud detection dataset like Partial Dependence Plots (PDP), Feature Importance, Shapley Additive Explanations (SHAP), SHAPASH, and Local Interpretable Model-Agnostic Explanations (LIME). After hyperparameter tuning, the experimental results showed that the Light GBM model performed better with 98.3% Accuracy, F1- score of 0.70, and AUC-ROC score of 0.96 compared with other models. The outcome of the research can be applied to design Proof of Concept (POC) as a front-end web application and deployed in the cloud using Python and Streamlit to make it accessible to non-AI expert consumers like business managers, domain experts, business analysts, and internal stakeholders for better decision-making and generating business value through real-time predictions.

Keywords: Explainable Artificial Intelligence; Human-centric AI; Machine Learning; Deep Neural Networks; Financial Transaction; Fraud Detection.

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1. Introduction

Artificial Intelligence (AI) has entered the business mainstream opening new opportunities and having the vast potential to solve the most complex business problems in multi-disciplinary fields. According to PwC's report, AI will be a $15.7 trillion opportunity by 2030 (PwC, 2020). During the Covid-19 pandemic, the use of online payments has skyrocketed, and the world is

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making an inexorable shift toward digital transformation by using more credit cards, debit cards, and mobile apps instead of cash in their daily life from being a convenience to being a necessity. As per UK Finance 2022, there were 284.1 million credit card transactions in February, which is 40% more than in February 2021 (UK Finance, 2022). As per Statista research in Fig. 1, with an annual payment processing volume of more than $10 trillion, Visa is undoubtedly the market leader among digital payment processors, followed by Apple Pay and Alipay, processing more than $6 trillion in transactions (Buchholz, 2022).

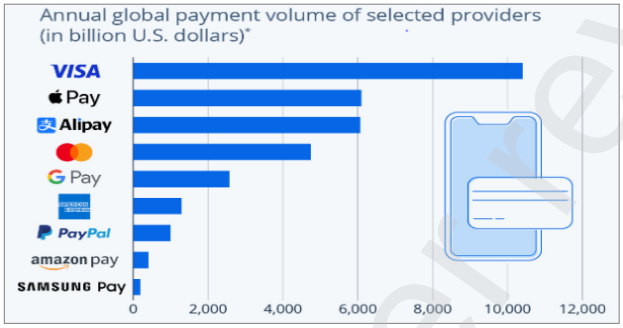

Fig. 1. Annual Global Digital Payments of Selected Credit Card Providers (Buchholz, 2022)

1.1 Impact of AI on Finance and Banking

Over the last several years have witnessed a rapid increase in the adoption of AI and Data Science in the finance and banking industry. The financial industry has benefited from introducing cutting-edge digital technologies like AI, creating a new discipline of FinTech (Julias, 2020). With the adoption of AI, FinTech uses automation and seamlessness of procedures to improve finance, sometimes without or with minimal human participation. The global banking industry is estimated to save $1 trillion by 2030 and $447 billion by 2023 due to the use of artificial intelligence (Digalaki, 2022).

Artificial intelligence (AI) is used in FinTech to create solutions specifically tailored to the problems that traditional finance and banking sectors, like Fraud Detection, Risk Management, Investment Management, Predictive Analytics, and Anti-Money Laundering. The banking sector is profiting more from AI technology within the financial services industry. Banks have made better business decisions by adopting AI to automate investment operations, improve customer relationship management, detect fraud more accurately, and cut operational expenses through Robotic Process Automation (RPA). Fig. 2 demonstrates how AI has revolutionized the banking and financial services industry in multidisciplinary areas.

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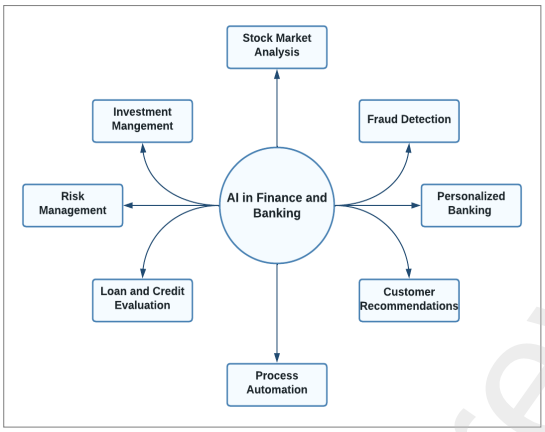

Fig. 2. Applications of AI in Finance and Banking (Image by Authors)

1.2 Financial Transaction Fraud Detection

Online payment systems are frequently utilized to complete financial transactions. As a result, businesses and entrepreneurs are adopting digital technology at an increasing rate. The digital payment systems that support online financial transactions are quicker than conventional rule-based transaction systems. While digitization holds great promise for improving business practices and growth, it also attracts fraudsters and cyber criminals who use their abilities to commit new financial frauds. According to UK Finance, fraud losses increased by 30%, and fraudsters have stolen £754 million from bank financial transactions (UK Finance, 2021). In the past, businesses have dealt with criminals who have gained unauthorized access to their assets to steal money. Today, it is possible to make financial transactions online by entering your credit card number and verifying your identity with a PIN (personal identification number). This has led to a significant problem for financial institutions that rely on digital card payments.

In the UK, the banking system avoided £1.4 billion of unauthorized fraud. However, criminals continued to take advantage of the pandemic last year by pretending to be several different organizations to commit fraud, and in 2021 they stole over £1.3 billion through both authorized and unauthorized fraud (UK Finance, 2022). More than 76% of credit card fraud losses in the UK were due to Card-not-Present (CNP), totalling £470.2 million. Additionally, financial firms and banks have been subject to regulatory fines totalling $321 billion over the past ten years for breaking laws about anti-terrorism financing, anti-fraud, and anti-money laundering (McKinsey & Company, 2019). Banks are attempting to find ways to enable quicker, more secure payments without degrading the customer experience. Due to the necessity to safeguard customer interests and legal repercussions, adopting AI-powered fraud detection systems has become vital for financial institutions and banks to stop online transaction fraud.

1.3 Rule-based vs. AI/ML-based Fraud Detection System

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Over the past few years, a movement has occurred in the finance and banking industry toward rule-based fraud detection systems supplemented by AI/ML-based fraud detection. Fig. 4 demonstrates the Traditional Rule-based vs. AI/ML-based Fraud Detection System. Current systems for detecting changes in behaviour and forecasting outcomes are based on pre-programmed rules. These rigid systems cannot adapt to a financial services sector that is highly disruptive and dynamic, which calls for a more adaptable platform to address fraud issues. Contrarily, AI-powered systems provide real-time insights, anomaly detection, and the detection of minute changes in patterns in massive and dynamic data sets.

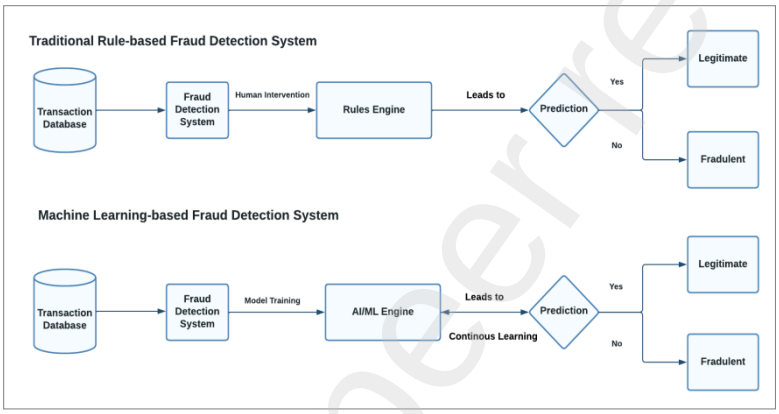

Fig. 3. Traditional Rule-based vs AI/ML-based Fraud Detection System (Image by Authors)

Table 1

Overview of Traditional Rule-based vs. AI-based Fraud Detection System

Rule-based Fraud	AI/ML-based Fraud
Detects fraud based on if-else rules, which are set manually by domain experts/analysts	Detects fraud automatically by ML algorithms without any human intervention
More time to process	Real-time processing
More verification methods are needed.	Less time is needed for verification methods.

1.4 The layout of the article

The remainder of the article is organized in the following manner. In Section 2, the background of AI and Explainable AI are covered. The related work on four domains, financial transaction fraud detection, ML, DNN, and XAI, is reviewed in Section 3. Section 4 represents the research framework, research methods of ML/DNN algorithms, and Explainable AI (XAI) methods with theoretical concepts, taxonomies, and precise analysis. The results and evaluation metrics, such as Precision, Recall, Accuracy, F1 score, AUC-ROC curve, and Confusion matrix

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to measure the performance of the fraud detection models implemented, are presented in Section 5. Finally, section 6 concludes the article with an indication of our future research works.

2. Background of AI and Explainable AI (XAI)

2.1. Artificial Intelligence (AI)

Artificial intelligence (AI) is the simulation of human intelligence by machines. The term "AI" can also refer to any machine that displays human cognitive abilities, such as learning and problem-solving (Frankenfield, 2022). AI can solve human-defined objectives by offering forecasts, suggestions, and other decision-support tools. In simpler terms, AI is a science concerned with building smart and intelligent machines.

2.2 Relation of AI with ML and DL

Nowadays, the terms Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL) are frequently used interchangeably to describe intelligent software or systems. Machine Learning is a subbranch of AI, is a subbranch of Deep Learning (DL). AI is the application of expertise gathered by humans to create systems or machines that perform tasks requiring reasoning and decision-making. ML is an automated process that learns from data to create models capable of performing specific tasks (Sarker et al., 2021). DL is a data-driven ML approach that involves multi-layer neural networks and computation. The term "deep" refers to multiple levels of processing in a data-driven model. (Sarker, 2021).

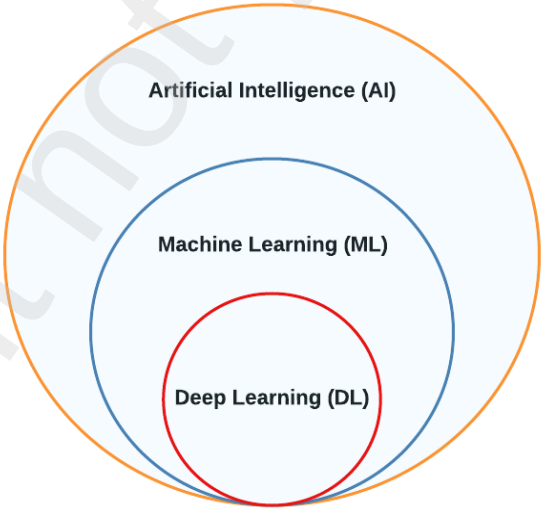

Fig. 4. Relation between AI, ML and DL (Image by Authors)

2.3 Explainable AI (XAI)

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An emerging field of AI termed Explainable AI (XAI) aims to transform complex "black-box" AI algorithms and system outputs into clearly comprehensible, reliable, or so-called "white-box" AI algorithms by relying on naturally understandable and straightforward models (Nicodeme, 2020). Without losing prediction accuracy, XAI makes black-box AI models more comprehensible, intuitive, and explainable to business decision-makers. (Deloitte, 2022). According to (Gunning, 2019), Explainable AI (XAI) will help users effectively trust the new generation of black-box AI systems using explainable and interpretable models. The type of explanation will change depending on the business decision-maker's role. For example, a business manager, fraud detection analyst, or domain expert, would need an explanation different from a model developer or data scientist who develops the AI fraud detection model. Hence, businesses depend upon AI more, and an explanation of black-box AI model outcomes is essential, particularly in how XAI-derived decisions generate business value and impact customers.

2.3.1 Traditional AI vs Explainable AI

Today's AI technology delivers a prediction or recommendation for businesses. However, non-AI expert users like business managers, domain experts, business analysts, and internal stakeholders cannot easily perceive how the decisions are achieved or why the model predicted this result. XAI, as indicated in Fig. 5 is a human-centric AI that uses an explainable model with an explanation interface where users can understand and trust AI-made decisions without compromising accuracy for better decision-making and generating business value.

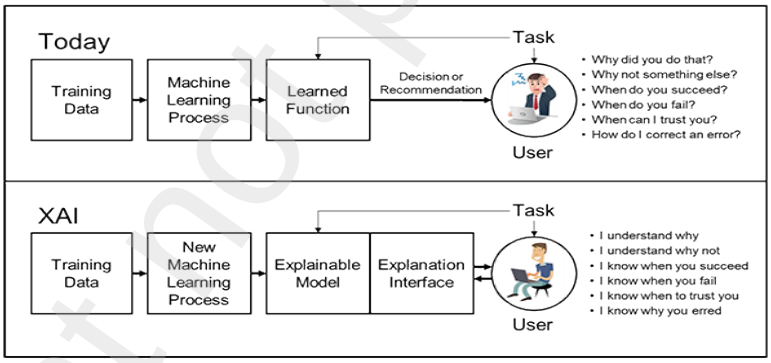

Fig. 5. Traditional AI vs XAI Framework (Turek, 2021)

2.3.2. Model Explainability vs. Model Accuracy

There is a wide range of models available today, from traditional ML Decision Trees to cutting-edge DNNs. With increased attention to model accuracy and explainability, as indicated in Fig. 6, XAI can assist AI developers in weighing these trade-offs more concretely and guide how to start bridging the complexity and intuitiveness gaps (Deloitte, 2022). While conventional

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ML models may be easier to understand, they frequently lack the accuracy and predictive power of more advanced models like DNNs.

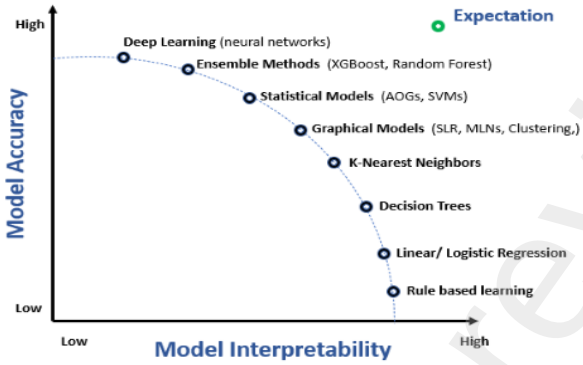

Fig. 6. Model Explainability vs. Model Accuracy (Arya AI, 2021)

2.3.3. Taxonomy of XAI Methods

2.3.3.1. Model-Agnostic vs. Model-Specific

Model-agnostic methods can describe any model because a specific model architecture does not restrict them. On the other hand, model-specific processes are restricted to a specific model architecture, demanding access to the model's data. (Patrício et al., 2022).

2.3.3.2. Global Interpretability vs. Local Interpretability

The patterns in the data, or the class features, that most significantly influenced the model's prediction are identified using global interpretability approaches. In contrast, local interpretability techniques aim to clarify why a model makes a particular prediction for an input.

2.3.3.3. Post-hoc vs Intrinsic

To understand the role that various features play in model prediction, post-hoc approaches often operate by perturbing certain portions of the data or mathematically calculating the role that various features play. Contrarily, intrinsic models, also known as interpretable models, are self-explanatory since they are designed to produce conclusions based on the internal model components and are thus understandable by humans.

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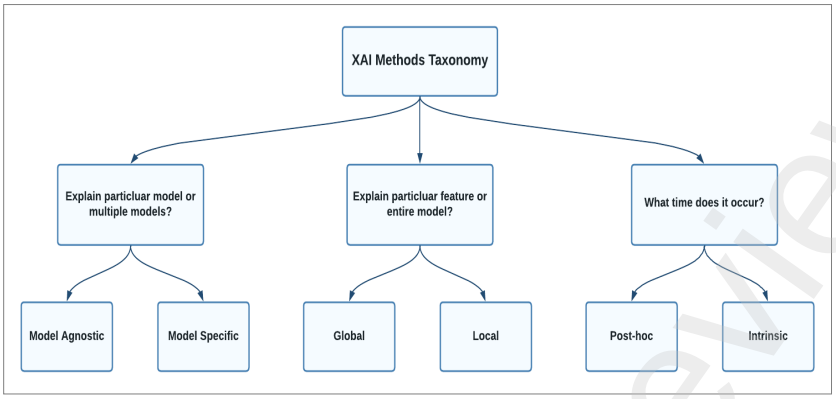

Fig. 7. Taxonomy of XAI Methods (Image by Authors)

3. Related Work

The Explainable AI-driven application for detecting financial transaction fraud is implemented across four domains: financial transaction fraud detection, ML, DNN, and XAI. A literature review was first required to get a general understanding of the state-of-the-art in fraud detection, which ML and DNN methods have been applied with good performance, as well as to identify XAI methods to support the business decision-makers in the field so that the direction of this project can be clear and define the inherent challenges and contributions. We chose references that were published after 2018 to identify the state-of-the-art ML, DNN, and XAI approaches used within the domain of financial transaction fraud detection and were sourced from the databases of Google Scholar, IEEE Xplore, Arxiv, Research Gate, Association for Computing Machinery (ACM) and Semantics Scholar. The paper screening process was conducted and screened by searching for the relevant keywords of this project in terms of title and abstract. In this research work, a literature review was needed to answer the following questions:

- What is the current state of financial transaction fraud detection using ML and DNN methods?
- What is Explainable AI (XAI), and how is it different from Traditional AI?
- Why are XAI explanations needed for the financial services and banking industry?
- How can non-AI expert users trust and interpret the outcomes of black-box AI models?

3.1. Literature Review Analysis

ML for fraud detection has been around for a while, and many researchers have created various combinations of different ML methods to increase fraud detection's accuracy. However, financial datasets share a similar problem: they are imbalanced because fraudulent transactions happen less frequently than regular transactions. (Wu and Wang, 2021).

3.2. Critical Analysis of Literature Review

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Table 2

Critical Analysis of Literature Review

Title	Citation	Research Problem	Results/Limitations
CC Fraud Detection - ML methods	(Varmedja et al., 2019)	Classify transactions as fraud or genuine by comparing ML models	Only ML Algorithms were used. Model Evaluation is done only with Accuracy.
CC Fraud Detection Based on ML & DL	(Najadat et al., 2020)	Tackle the problem of cc fraud using ML & DL	BiLSTM & BiGRU achieved 91.37% Accuracy. Model Evaluation is done only with Accuracy.
DL Methods for CC Fraud Detection	(Nguyen et al., 2020)	Performance of the DL methods against ML models for the CC fraud detection	LSTM with 50 blocks was the one on top with an F1-Score of 84.85%. No Hyperparameter tuning was applied to DL methods.
CC Fraud Detection using ML and DL Techniques	(Azhan and Meraj, 2020)	How to identify fraudsters utilizing past errors leveraging machine learning and neural networks.	Compared to shallow neural networks, machine learning techniques have proven to be more adept at addressing class imbalance.
An Efficient Real-Time Model for CC Fraud Detection Based On DL	(Abakarim et al., 2018)	Classify real-time CC transactions as legitimate or fraudulent	Deep NN Autoencoder has the best F1 score. The state-of-art DL methods like ANN and CNN still needed to be implemented.
CC Fraud Detection Using State-of-the-Art ML and DL Algorithms	(Alarfaj et al., 2022)	DL algorithms are used to reduce the false negative rate.	The most accurate approach is CNN, which has 20 layers and a 99.72% accuracy rate. Modern DL techniques weren't used.
CC Fraud Detection Using Artificial Neural Network	(RB and K R, 2021)	Use ML and DL algorithms to distinguish between fraudulent and legitimate transactions.	ANN predicts higher accuracy than SVM and KNN algorithms. Only three algorithms were

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			compared based on the accuracy metric.
Evaluation of DNN for Reduction of CC Fraud Alerts	(San Miguel Carrasco and Sicilia-Urbán, 2020)	The number of notifications is decreased by accurately discriminating well-known false positives using DNN.	Triggered warnings were reduced by 35.16%, while the remaining alerts contained 91.79% fraud cases.
Explainable ML for Fraud Detection	(Psychoula et al., 2021)	Exploring explainability methods for real-time fraud detection.	SHAP provides more trustworthy justifications, although LIME is quicker.
Explainable AI methods for CC fraud detection: Evaluation of LIME and SHAP through a User Study	(Ji, 2021)	It is researched how to make a credit card fraud detection support system understandable to users.	LIME has an advantage over SHAP in terms of explainability.
Explainable AI (XAI): Concepts, Taxonomies, Opportunities & Challenges toward Responsible AI	(Arrieta et al., 2019)	Give XAI a comprehensive taxonomy that serves as a source of information.	Summarised previous efforts made to define XML. There is a need for a proper understanding of the XAI techniques.

3.3. Summary of Related Work

In this section, we examined the current state of financial transaction fraud detection with ML, DNN and XAI methods that have been used for implementing fraud detection models.

- ML methods like Decision Tree, Logistic Regression, and Random Forest have been highly discussed and performed well in past research, but ensemble methods like Light GBM and XGBoost were less examined and applied. Hence, we will apply Light GBM and XGBoost.
- DNN methods like ANN and CNN have been significantly less applied and have satisfactory performance in fraud detection cases. To construct an XAI-driven explainable interface, a blend of five ML algorithms and two DNN algorithms will be evaluated to determine which approach performs best.
- There are many different explanations and approaches for fraud detection algorithms for the XAI methodologies. Due to time and space constraints, the five XAI methods selected for this project are Feature Importance, PDP, SHAP, LIME, and SHAPASH.

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4. Experiments

In this work, we have covered the research framework, research methods of ML/DNN algorithms, and Explainable AI (XAI) methods with theoretical concepts, taxonomies, and precise analysis to highlight the strength and weaknesses of the applied techniques to support the achievement of all the objectives.

4.1 Research Design Process

The AI research life cycle includes different processes required to develop an AI solution to address business problems like financial transaction fraud detection. The CRISP-DM method is the most effective way to describe such a process for deriving insights from data. We will use the industry standard CRISP-DM process to implement the financial transaction fraud detection model for this research study. CRISP-DM, which stands for Cross Industry Standard Process for Data Mining, is the most famous work approach involving data mining, data analytics, and Artificial Intelligence (AI) (Hotz, 2018).

4.2 Proposed Design Process

The industry-standard CRISP-DM procedure created for the research is shown in Fig. 8. This section will cover the proposed design framework that draws inspiration from the CRISP-DM to achieve the goals of the research framework (Hotz, 2018). Model explainability is one stage that is involved in the design process. Business organizations may better comprehend and manage their AI systems with the aid of explainable AI. A human-centric AI with model explainability is one of the fundamental elements for all AI laws, and it de-risks the adoption of AI. Therefore, to guarantee that AI project gives business organizations the necessary benefits, ensure the AI solution is explainable. A more inclusive, equitable, and morally sound future can be achieved with XAI. XAI can prevent consumers from accepting recommendations at face value by giving them insight into the machine's decision-making process and shifting the burden of proof back onto the human (Poria, 2022).

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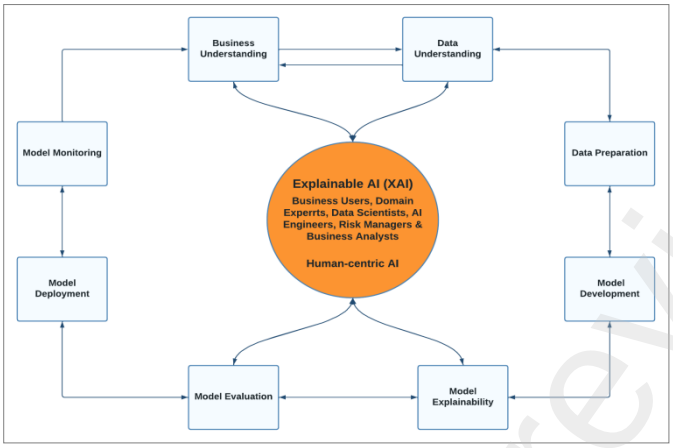

Fig.8. Proposed Design Process (Image by Authors)

4.3 Research Framework

Fig. 8 presents the proposed research design process, and the characteristic features of XAI-driven fraud detection have been proposed. Based on the research problem, the main research objective has been formulated to implement an "Explainable AI (XAI)-driven Financial Transaction Fraud Detection using Machine Learning and Deep Neural Networks".

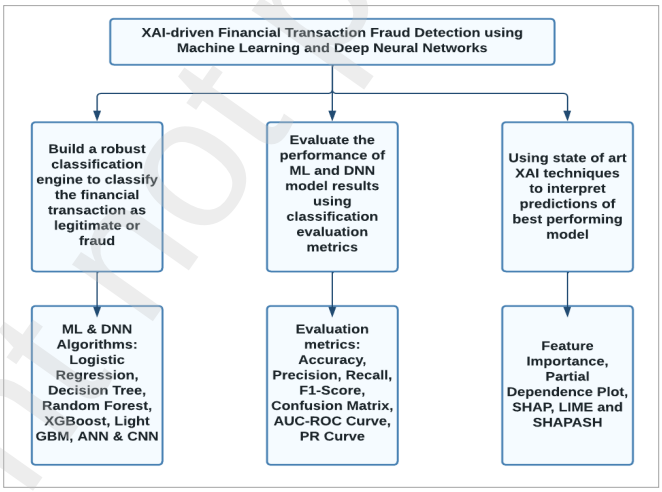

Fig.9. Overview of Research Framework (Image by Authors)

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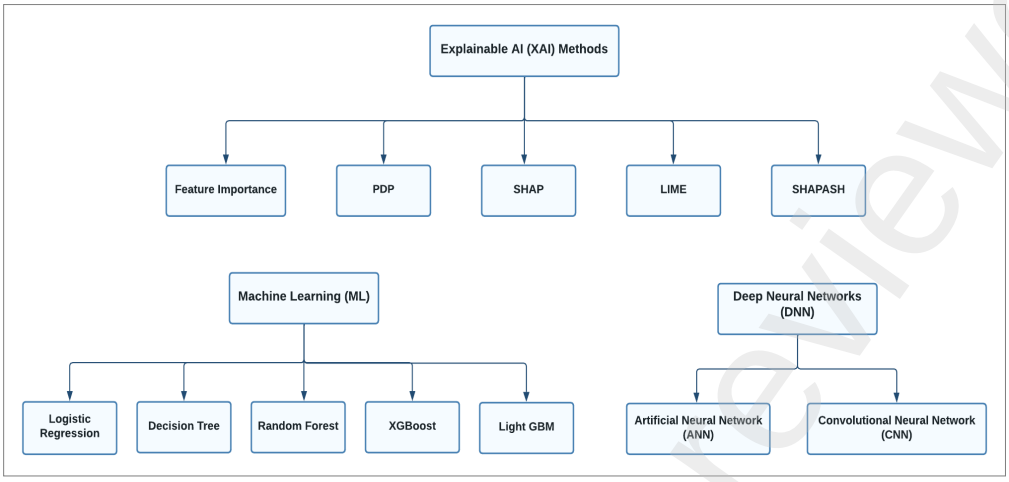

Fig.10. Overview of Research Methods - ML, DNN and XAI (Image by Authors)

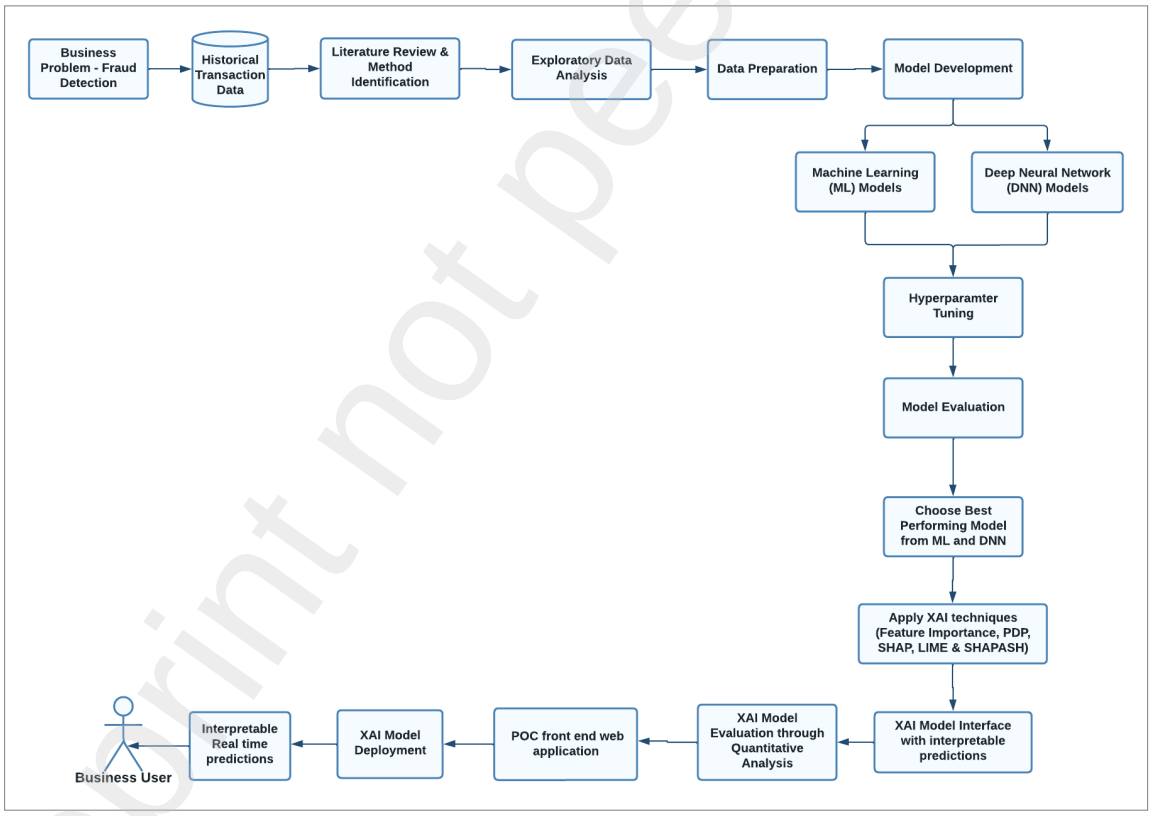

Fig.11. Overview of Research Workflow (Image by Authors)

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4.4 Research Methods – XAI

In this section, Fig. 10 shows the methodology employed to accomplish the research objectives formulated in the research framework has been discussed. XAI Methods are Feature Importance Plots, Partial Dependence Plots (PDP), SHAP, LIME, and SHAPASH will be introduced.

4.5 Research Workflow

In this section, Fig. 11 depicts an overview of the research workflow demonstrated to accomplish the aim and objectives formulated in the research framework have been discussed.

4.6 Data Preparation

Here, we'll examine how to prepare data for feature engineering and exploratory data analysis on a dataset for creating and evaluating fraud detection models.

4.6.1. Dataset Description

Here, we have chosen a dataset of genuine financial transactions provided by Vesta Corporation. More than $18 billion in transactions are now guaranteed yearly by Vesta. The data was gathered from actual e-commerce transactions by Vesta, the top payment service provider in the world, and the IEEE Computational Intelligence Society (IEEE-CIS), an organization that works in a variety of AI and machine learning fields, to identify the most effective fraud prevention techniques for the banking and financial sectors. This dataset is available in the link IEEE-CIS Fraud Detection Dataset. Data is divided into two 'identify' and 'transaction' files, which are connected by a transaction id.

- **Transaction.csv:** Contains transactional information to be used in model training. The transaction dataset has 590540 rows/data samples and 394 columns/features.
- **Identity.csv:** Contains information about the identity of the payer and the merchant between whom the transaction was made. It has 144233 rows/data samples and 41 columns/features.

4.6.2. Optimisation of Memory and Run-time

Real-world AI solutions must be delivered in a way that is both sustainably efficient at run-time and memory. Due to an extensive dataset in this research, we have optimized both run time and memory by creating a function that iterates through all features which occupy more memory than what is needed to store them. It assigns suitable datatype based on the values present in a feature. Reducing memory usage by changing data type will speed up the computations.

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Code for Memory and Run-time Optimisation

Certain features occupy more memory than what is needed to store them. Reducing the memory usage by changing data type will speed up the computations. As indicated in Fig. 12, the following code is implemented for Memory and Run-time Optimisation, where Table 3 summarises the memory and run-time optimisation.

```
# Create reduce memory usage function
def reduce_mem_usage(df, verbose=True):

  #define list of datatypes that are required to optimise
    numerics = ['int16', 'int32', 'int64', 'float16', 'float32', 'float64']

    #Initialise variable 'start_mem' to store initial memory occupied by dataframe
    start_mem = df.memory_usage(deep=True).sum() / 1024**2

    #Iterate through all columns in dataframe
    for col in df.columns:
        col_type = df[col].dtypes

        #Check datatype in list 'numerics'
        if col_type in numerics:
            c_min = df[col].min()
            c_max = df[col].max()

            #Compare the memory usage of each column and assign the necessary datatype

            if str(col_type)[:3] == 'int':
                if c_min > np.iinfo(np.int8).min and c_max < np.iinfo(np.int8).max:
                    df[col] = df[col].astype(np.int8)
                elif c_min > np.iinfo(np.int16).min and c_max < np.iinfo(np.int16).max:
                    df[col] = df[col].astype(np.int16)
                elif c_min > np.iinfo(np.int32).min and c_max < np.iinfo(np.int32).max:
                    df[col] = df[col].astype(np.int32)
                elif c_min > np.iinfo(np.int64).min and c_max < np.iinfo(np.int64).max:
                   df[col] = df[col].astype(np.int64)
            else:
                if c_min > np.finfo(np.float16).min and c_max < np.finfo(np.float16).max:
                    df[col] = df[col].astype(np.float16)
                elif c_min > np.finfo(np.float32).min and c_max < np.finfo(np.float32).max:
                    df[col] = df[col].astype(np.float32)
```

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```
            else:
                df[col] = df[col].astype(np.float64)

    #Store the Optimised memory used by dataframe in 'end_mem' variable
    end_mem = df.memory_usage(deep=True).sum() / 1024**2

    if verbose:
      print('Memory usage decreased and optimised to {:5.2f} Mb
          ({:.1f}% reduction)'.format(end_mem, 100 * (start_mem - end_mem) / start_mem))
  return df
```

Fig.12. Code for Memory and Run-time Optimisation (Image by Authors)

Table 2

Summary of Memory and Run-time Optimisation

Dataset	Before Memory (MB)	After Memory (MB)	Before Run-time	After Run-time
Transaction	2100.70	867.33	28 Sec	11.7 Sec
Identity	157.63	138.24	741 Mill sec	353 Mill Sec

4.6.3. Exploratory Data Analysis (EDA)

In this research, we have used Exploratory Data Analysis (EDA) approaches to extract insights from the data set to know which features have contributed more to predicting the target attributes by doing Data Analysis and Visualization using Python, Pandas, NumPy, Matplotlib, and Seaborn as indicated in Fig. 13, Fig. 14, Fig. 15, and Fig. 16.

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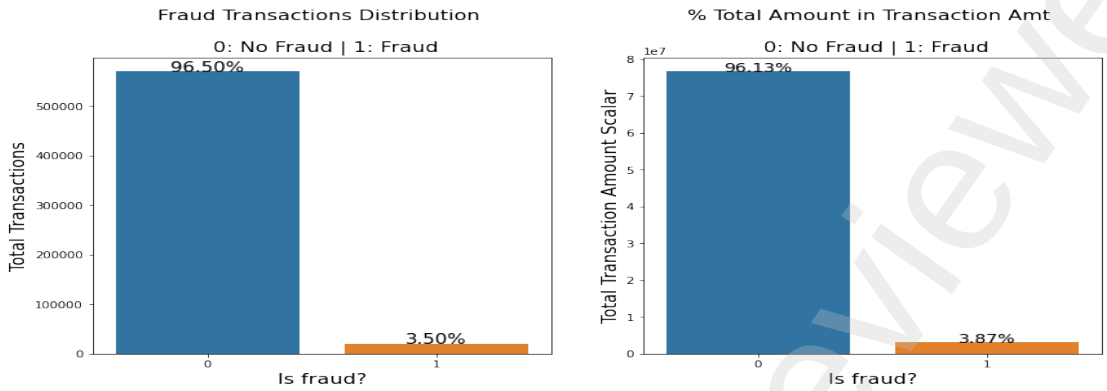

Fig. 13. Distribution of Financial Transactions

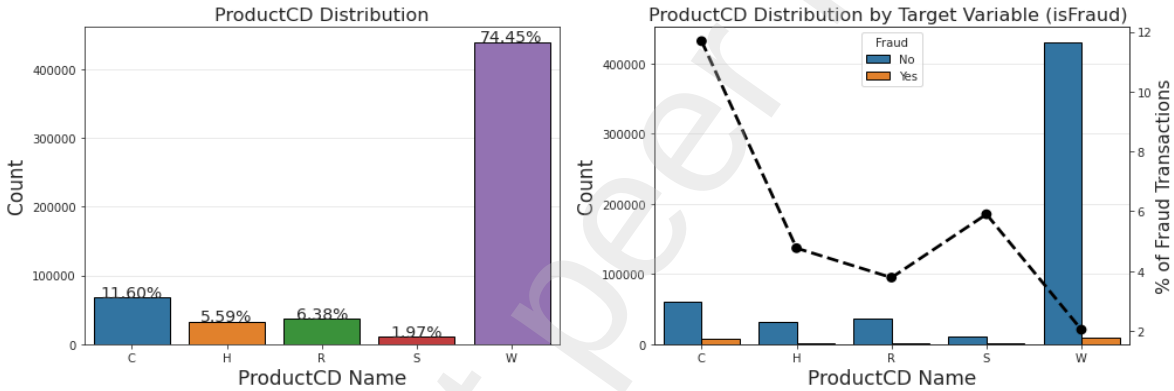

Fig. 14. Distribution of Product Code

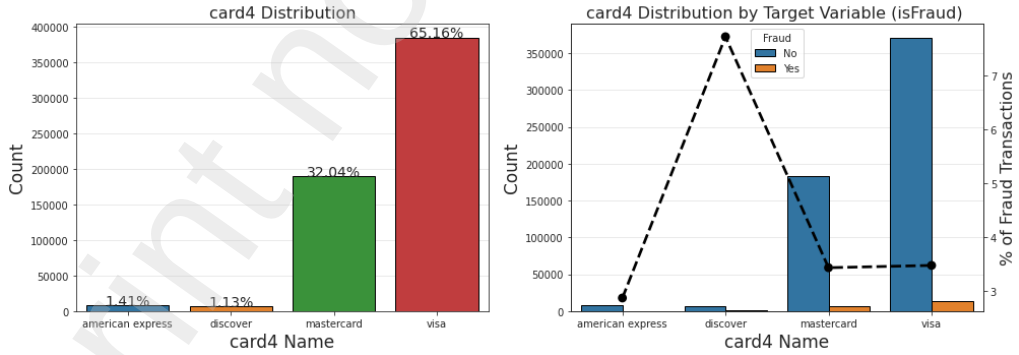

Fig. 15. Distribution of Card4

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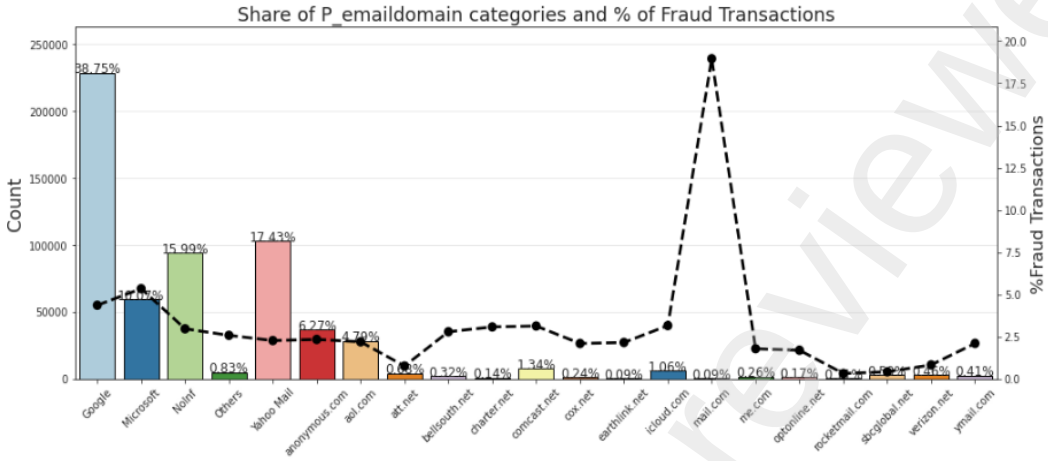

Fig. 16. Distribution of Purchaser Email Domain

EDA Inferences:

1. The target variable 'isFraud' is highly imbalanced, with 3.5% of transactions as fraud and 3.87% of transaction amounts as fraud. The average legitimate transaction Amount is 134.5 USD, and the fraud transaction Amount is 149.2 USD. Hence, fraudulent transactions occur when the transaction amount is higher.
2. 74.5% of the transactions are for Product Code 'W' and 11.6% for Product Code 'C'. The fraud Transaction Rate is Maximum for Product Category 'C' and minimum for 'W'.
3. About 97% of transactions are from Mastercard (32%), and Visa (65%) and the fraud transaction rate is highest for Discover cards (7.72%).
4. About 38.78% of transactions are from Google and 17.43% from Yahoo Mail. The highest fraud % rate is from mail.com (18.96%), Microsoft (5.32%), Google (4.34%) and Yahoo (2.25%).

4.6.4. Data Pre-processing and Feature Engineering

4.6.4.1. Principal Component Analysis (PCA)

PCA is a technique for reducing the dimensionality of these datasets without losing information and enhancing interpretability. In this research project, we have implemented PCA to handle this large dataset's high dimensionality. Applying the PCA to this dataset reduced the dimensionality of features to 537 features.

4.6.4.2. *Feature Encoding*

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Encoding means transforming data from one datatype to another. Most machine learning algorithms cannot accept category values until we convert them into numerical numbers. In this section, we applied both frequency and label encoding to categorical features.

- **Frequency Encoding:** The frequency of the categories can be used as labels. When the frequency is somewhat correlated with the target variable, it aids the model's comprehension and weighting decisions, which depend on the data.

- **Label Encoding:** For machines to read labels, they should be transformed into a numeric representation. This approach assigns a unique number to every label based on alphabetical order. ML algorithms can then decide how labels can be used more efficiently.

4.7 Implementation

In the previous section, we have pre-processed data for modelling. Now, we will develop a robust classification engine to predict a financial transaction as fraudulent or legitimate by applying five Machine Learning Algorithms, two Deep Neural Network Algorithms, hyperparameter tuning, and XAI methods.

4.7.1. Machine Learning Modelling

4.7.1.1. Splitting the Dataset

The dataset is split into a training set with 70% random observations and a test set with 30% random observations. Experimenting with both train/test splits showed that the 70/30 split appears to have a better representation of both positive and negative classes in training and testing datasets than an 80/20 split. The model will be tested on a test set once it has been trained on a training set. When splitting the data, we specified a random seed (random number) to ensure the same data was split every time.

4.7.1.2. Model Training

A machine learning algorithm must be trained with data to find and learn the best values for all associated variables. Five machine learning algorithms like Decision Tree, Logistic Regression, Random Forest, Light GBM, and XGBoost proposed in the research framework are utilized for training the ML models to identify fraud, and the testing dataset is used to evaluate the performance of the trained ML models for classification.

4.7.2. Deep Neural Networks Modelling

4.7.2.1. Splitting the Dataset

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The dataset is split into a training set containing 70% random observations and a test set with 30% random observations. Experimenting with both train/test splits showed that the 70/30 split appears to better represent both positive and negative classes in training and testing datasets than an 80/20 split. The Deep Neural Network models will be trained on a training set, and their performance will be assessed on a test set.

4.7.2.2. Model Training

The implementation of two Deep Neural Network Algorithms as proposed in the research framework using ANN and CNN.

4.7.3 Hyperparameter Tuning

4.7.3.1. Hyperparameter Tuning

The process of selecting the right combination of hyperparameters for the ML/DL model is known as hyperparameter tuning. The hyperparameters control the structure and training procedure of a model. These variables remain constant throughout the training phase and directly impact the model's performance. Selecting the best set of hyperparameters for an algorithm is the process of tuning. Hyperparameters are identified during tuning, whereas model parameters are obtained by fitting a model to the data. A loss function is minimized throughout the tuning phase to find the ideal hyperparameters.

After, model evaluation for ML and DNN algorithms using evaluation metrics like Precision, Recall, Accuracy, AUC-ROC score, and F1-score, the Light GBM classifier algorithm performed better with an Accuracy of 97.69%, AUC-ROC score of 0.93, and F1-score 0.56 when compared with other algorithms. Hence, for hyperparameter tuning, we will tune different parameters for the Light GBM classifier to predict fraudulent or legitimate transactions accurately and effectively.

4.7.3.2. Grid Search Cross Validation (GridSearchCV)

Grid Search is used to determine the efficiency of each combination of the hyperparameters, and the combination with the best performance is then selected. To train the model, cross-validation is used. The cross-validation process divides the train data into the validation and train data. To analyse the best hyperparameters and create an optimized model, Grid Search combined with the cross-validation technique requires significant overtime work.

Table 4

Light GBM Best Hyperparameter Values

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Hyperparameter	Hyperparameter Values	Best Hyperparameter Values
n_estimator	[100,500,200]	100
num_leaves	[128,256]	256
max_depth	[8,10,20]	20
learning_rate	[0.05,0.1]	0.1

4.7.4 Explainable AI (XAI) Demonstration

This section discusses the methodology employed to accomplish the research objectives formulated in the research framework. It briefly describes the state-of-art five XAI methods implemented in this project to develop the XAI-driven financial transaction fraud detection model. XAI Methods are Partial Dependence Plots (PDP), Feature Importance Plot, SHAP, LIME, and SHAPASH will be implemented and demonstrated.

4.7.4.1. Feature Importance

The term "feature importance" refers to a group of methods for assigning scores to input variables in a predictive model describing each feature's relative weighting in a prediction. To calculate feature importance scores, regression, and classification tasks, which involve forecasting a numeric value or a class label, can be used (Brownlee, 2020). As indicated, Fig. 17 shows how each aspect of a data point influences a particular prediction.

4.7.4.2. Partial Dependence Plot (PDP)

Plots of partial dependence indicated in Fig. 18 shows the relationship between the response and a subset of the target features, typically one or two. Due to the limitations of human perception, the input characteristics of interest could be few. Hence the most significant features are typically chosen as the input features of interest (scikit-learn, 2022).

4.7.4.3. SHAP (Shapley Additive Explanations)

Lundberg and Lee created a method known as SHAP (Shapley Additive Explanations) to explain individual predictions. The game's theoretically ideal Shapley values are the foundation for SHAP (Molnar, 2022). Tabular and picture data can be interpreted using SHAP, a local model-independent technique.

When using tabular data, the explanation is provided as relevance scores for each attribute. Kernel SHAP, Deep SHAP, and Tree SHAP are a few versions of the SHAP approach suggested to better accurately approximate Shapley values (Lundberg et al., 2019). Various versions of the SHAP Plot are demonstrated here where Fig. 19 indicates the SHAP Summary Plot, Fig. 20 demonstrates SHAP Dependence Plot, and Fig. 21 illustrates SHAP Force Plot.

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4.7.4.4. LIME (Local Interpretable Model-Agnostic Explanations)

LIME is a post-hoc, model-agnostic method that provides local explanations and, as its name suggests, can explain any black-box model. LIME's goal is to imitate the complex model with an understandable model locally, also called a local surrogate model. As a result, a specific occurrence is locally described using a straightforward interpretable model, as indicated in Fig. 22 and Fig. 23, based on the prediction, such as decision trees and the random forest.

4.7.4.5. SHAPASH

A Python package called SHAPASH attempts to make machine learning accessible to everyone by making it interpretable and understandable. We can create various visualizations as indicated in Fig. 24 and 25 to show clear labels to everyone (SHAPASH, 2022).

With the help of the Python module SHAPASH, we can analyze and comprehend machine learning. It offers a variety of visualizations with clear, universally understood labels. Data scientists can simply communicate their findings and understand their models. End users can understand a model's decisions by summarizing the most essential features.

This study's first phase of research on developing a robust classification engine to classify whether financial transaction is fraudulent or legitimate using ML/DNN algorithms has been accomplished. It has been concluded that the Light GBM model provided high performance in fraud detection tasks. Among all the models developed for fraud detection, the Light GBM algorithm with optimal hyperparameters provides the best classification performance in improving the false-negative rate. Further, Explainable AI approaches proposed have been demonstrated using the best-performing model Light GBM obtained after hyperparameter tuning.

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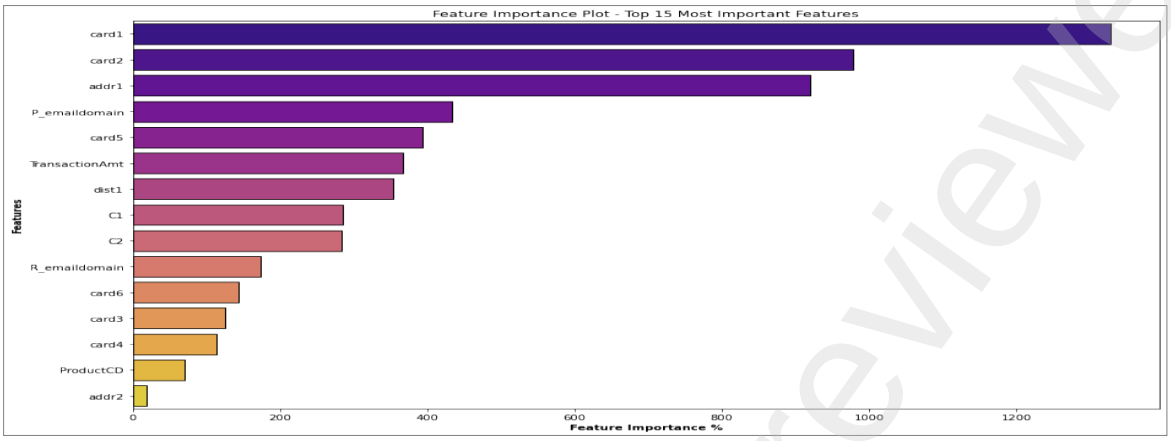

Fig. 17. Feature Importance Plot

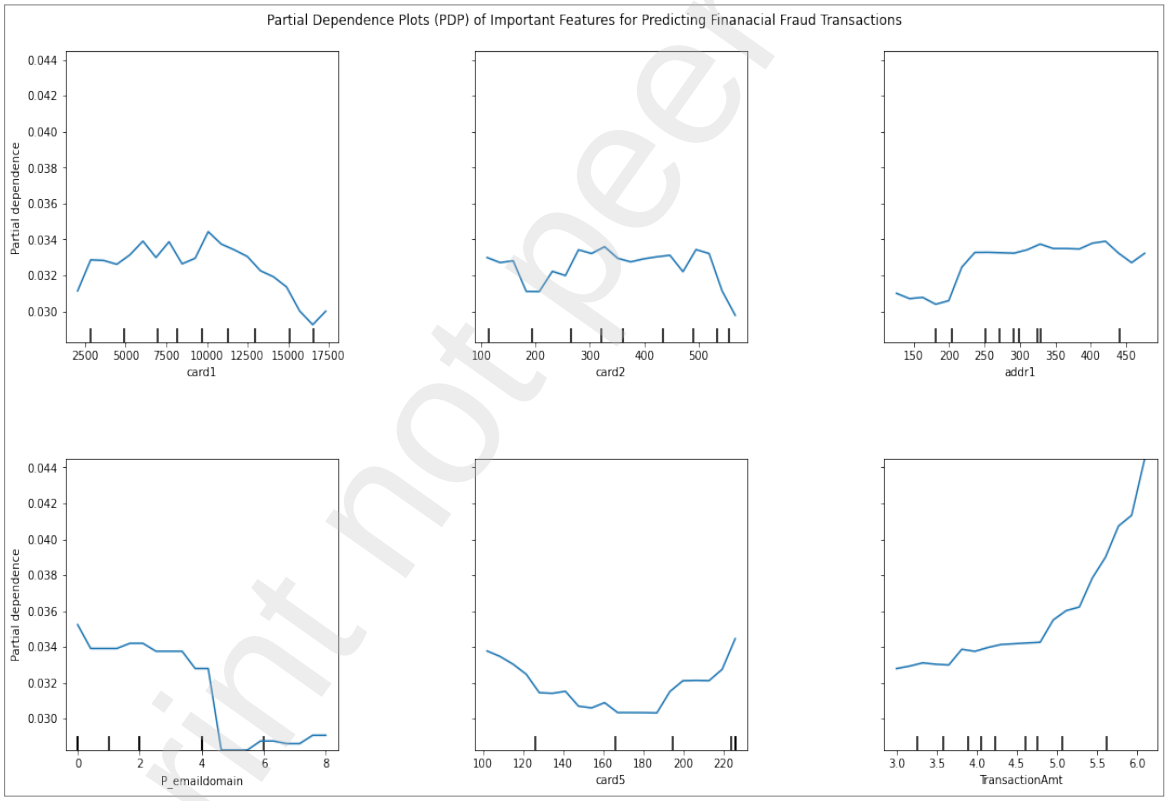

Fig.18. Partial Dependence Plot

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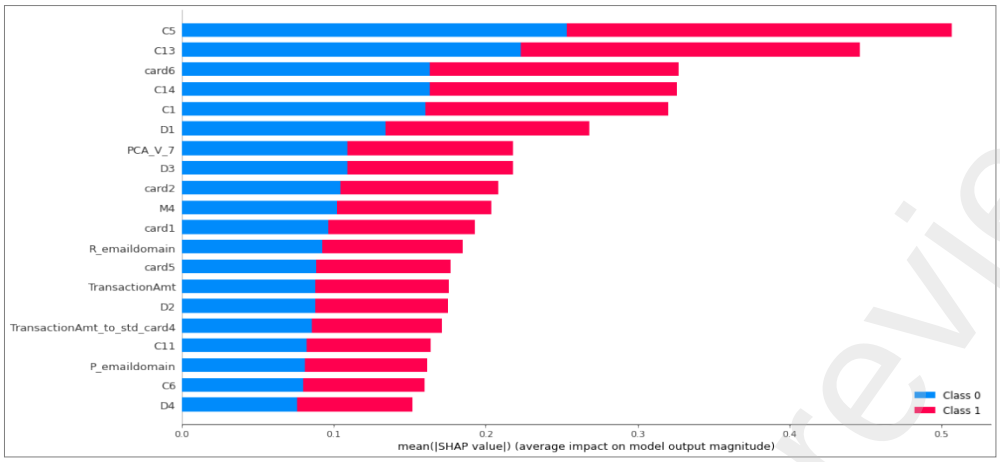

Fig.19. SHAP Summary Plot

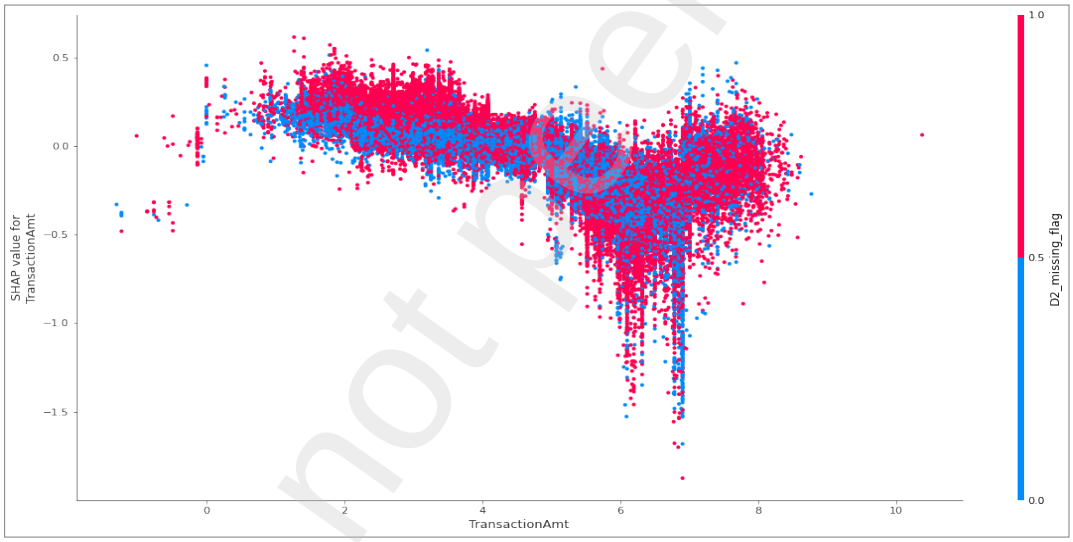

Fig.20. SHAP Dependence Plot

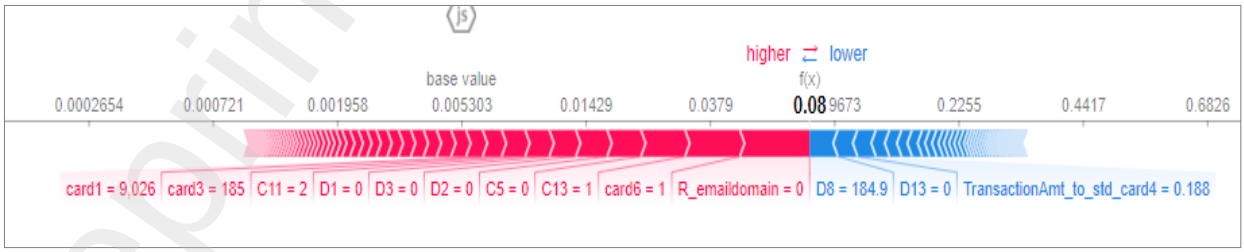

Fig. 21. SHAP Force Plot

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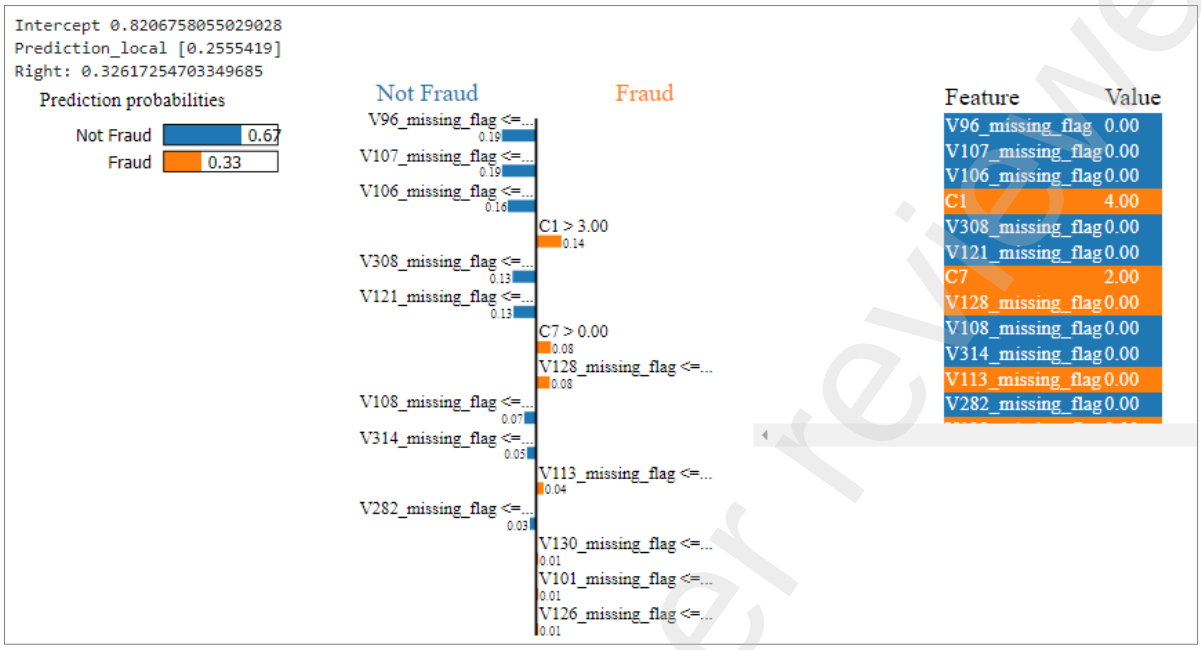

Fig. 22. LIME Plot

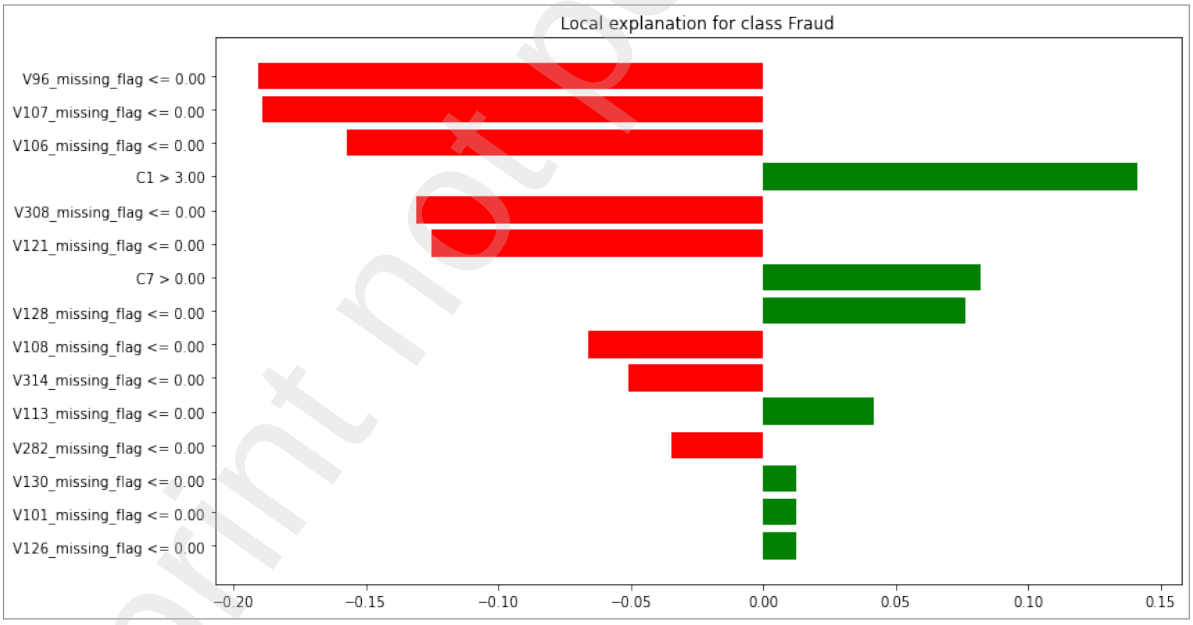

Fig. 23. LIME Local Explanation Plot

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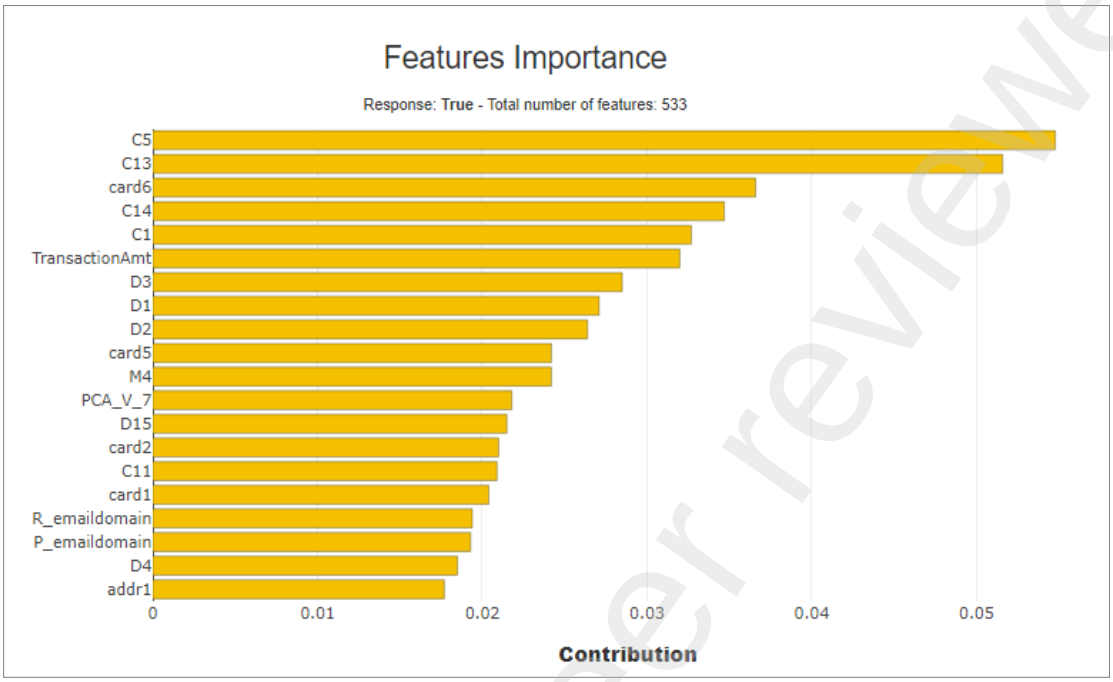

Fig. 24. SHAPASH Feature Importance Plot

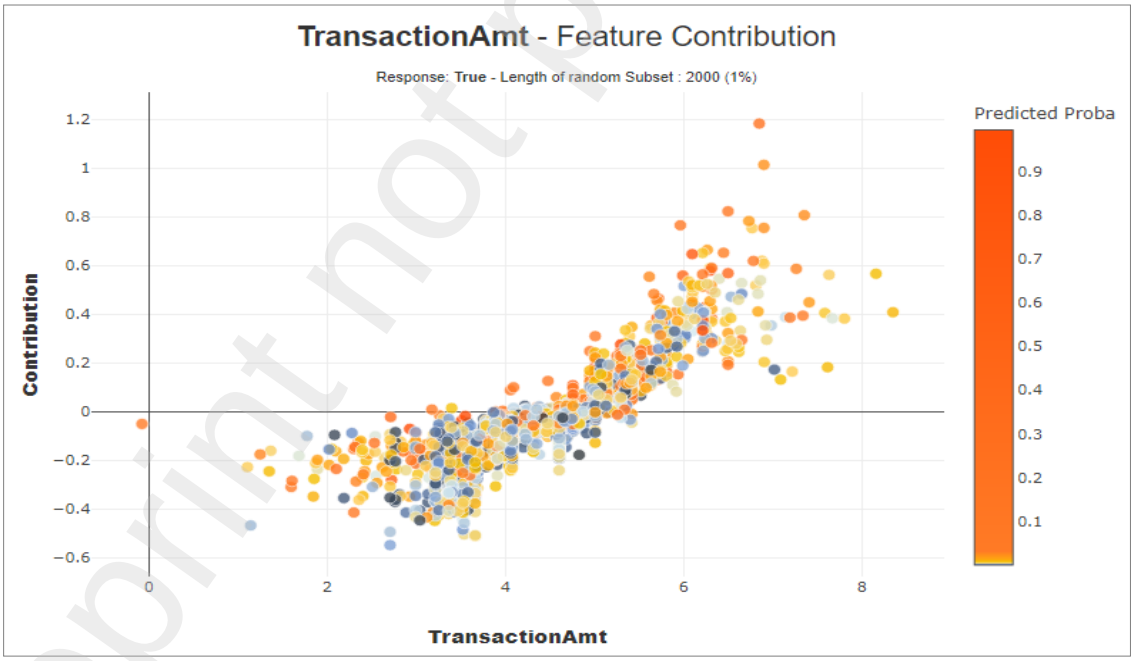

Fig. 25. SHAPASH Contribution Plot

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5 Results

Evaluation metrics like Precision, Recall, Accuracy, F1 score, AUC-ROC curve, and Confusion matrix have been utilised in this section to assess how well the ML/DNN fraud detection models performed at classifying data.

5.1 Classification Evaluation Metrics

In machine learning, the model's generalizability is verified after it has been trained using training data. In simpler terms, we examine the model's performance when put to the test on fictitious data. So how do we evaluate the model's effectiveness? Whether it is a regression problem or a classification challenge, we use evaluation metrics to determine how well the model works.

5.1.1. Confusion Matrix

A N x N matrix, N is the total number of class labels, is used to characterize a model's overall performance when applied to a dataset. We have a 2x2 confusion matrix for binary classification, as shown in the diagram in Fig. 26.

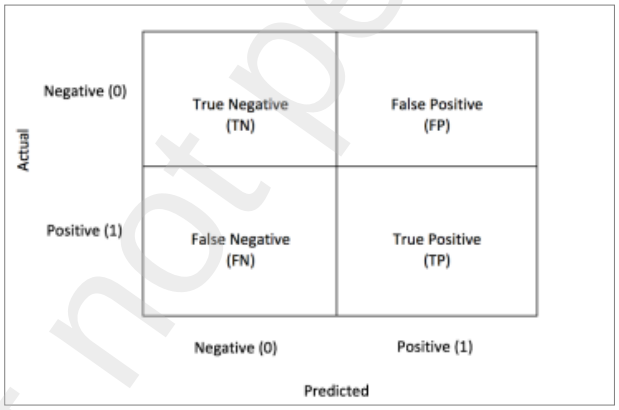

Fig. 26. Confusion Matrix (Image by Authors)

A confusion matrix is made up of statistics such as True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN), which are calculated using a combination of actual and predicted values.

- True Positive (TP): When both the actual and predicted values are positive.
- False Positive (FP): When the predicted value is positive, but the actual value is negative.
- True Negative (TN): When both the predicted and actual values are negative.
- False Negative (FN): When the predicted value is negative, the actual value is positive.

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5.1.2. Precision

It is the fraction of true positives compared to both true and false positives. Simply put, the precision measures the proportion of real positive cases discovered.

$$Precision = \frac{TP}{TP + FP}$$

5.1.3. Recall

It is the fraction of true positives over actual positive cases. In simple terms, recall is how many true positives are out of all the true positive cases.

$$Recall = \frac{TP}{TP + FN}$$

5.1.4. F1-score

It depicts the harmonic mean of recall and precision. It is the value can be between 0 and 1, with 1 being the best and 0 being the worst.

$$F_1 = \frac{2 * (Precision * Recall)}{Precision + Recall}$$

5.1.5. Area Under Receiver Operating Characteristic (AUC-ROC) Curve

The AUC-ROC curve is one of the most often utilized evaluation metrics in predictive analysis. It demonstrates how well a model works when used with different probabilities. 0.5 is chosen as the probability threshold for the classification problem.

5.1.6. Precision-Recall (PR) Curve

The PR curve is a helpful evaluation statistic to assess a predictive model's efficacy in unbalanced class distribution. It is based on the two-evaluation metrics, recall, and precision, and the graphic shows their values at different probability levels.

5.2 Model Evaluation Summary

Based on the analysis of the model performance metrics, below are the inferences,

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- With various degrees of performance, ML and DNN models can both identify fraud in financial transaction data. Evaluating the performance measures provides a thorough understanding of each model's advantages and disadvantages.

- To identify fraud transactions, boosting tree algorithms like Light GBM and XGBoost outperform ensemble tree-based models like Decision Tree and Random Forest and linear models like Logistic Regression.

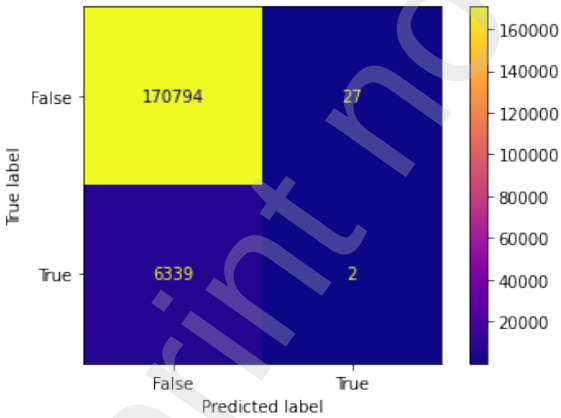
- Considering all the evaluation measures, light GBM models outperform all other ML and DNN models with 98.27% Accuracy, AUC-ROC score of 0.96, and F1-score of 0.70.

- In this research, Light GBM, and XGBoost outperform ANN and CNN-based fraud detection models in terms of performance. This is understandable, given the research demonstrating the superior performance of tree-based ensemble models over DNN models on tabular datasets.

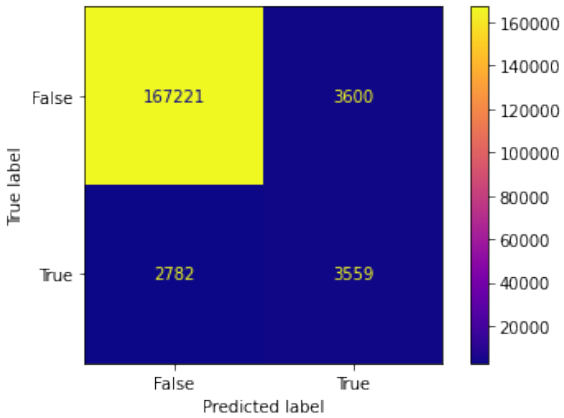
5.3 Machine Learning Model Evaluation

The results obtained through the evaluation of implemented machine learning models are presented through the figures, where Fig. 27 indicates the Confusion Matrix, Fig. 28 presents AUC - ROC Curve, Fig. 29 shows Precision-Recall (PR) Curve, and Table 5 presents the machine learning model evaluation summary.

5.3.1. Confusion Matrix



Logistic Regression



Decision Tree

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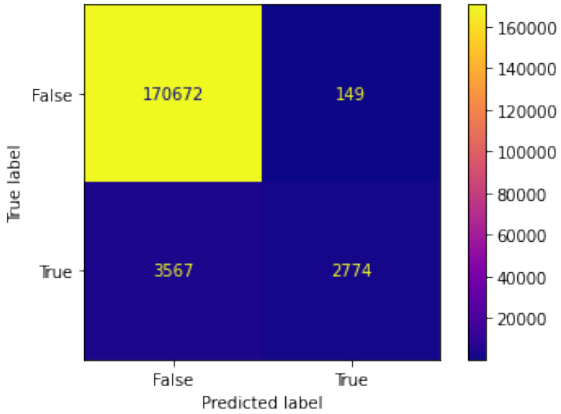
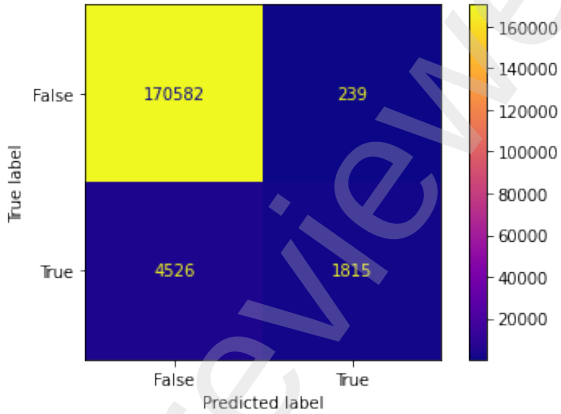
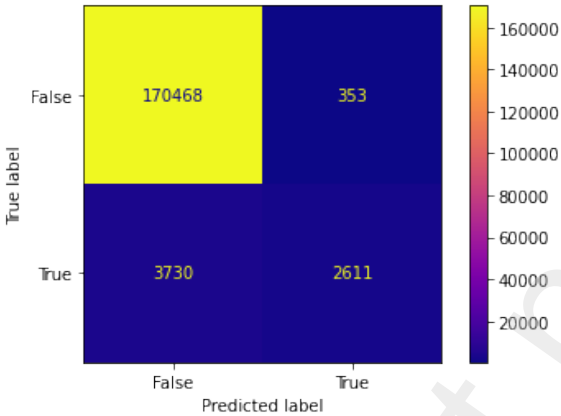


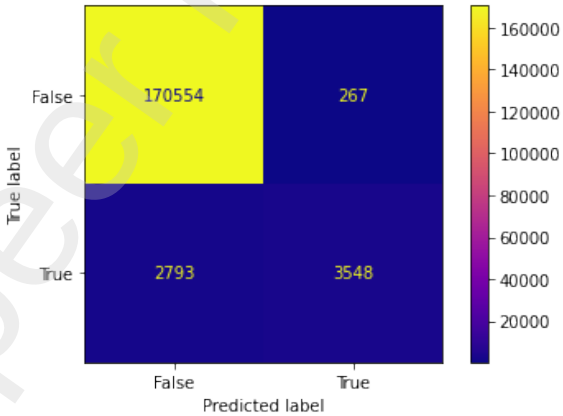
Random Forest



XGBoost



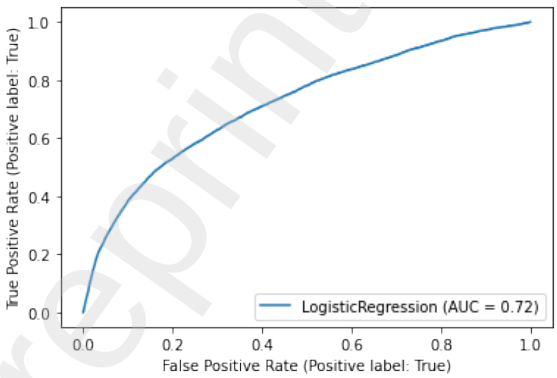
Light GBM Without Hyperparameter Tuning

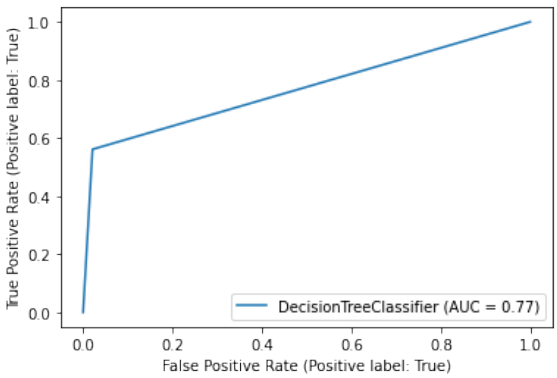


Light GBM With Hyperparameter Tuning

Fig. 27. Confusion Matrix

5.3. 2. AUC – ROC Curve





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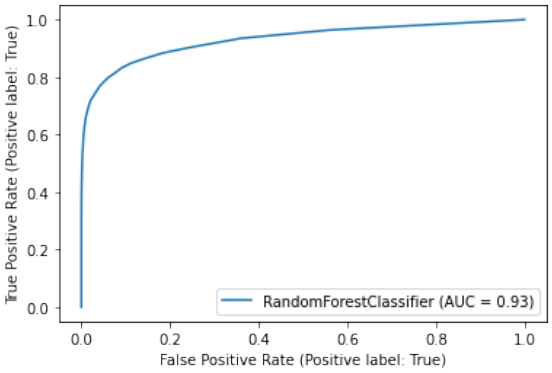
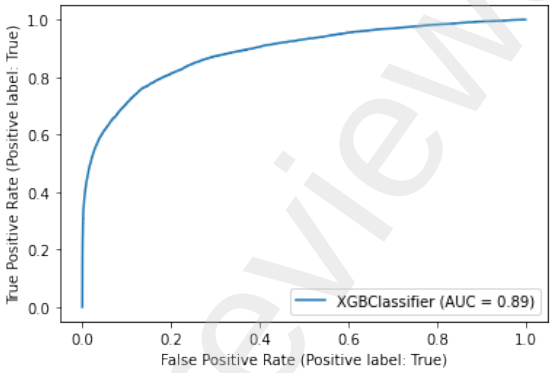
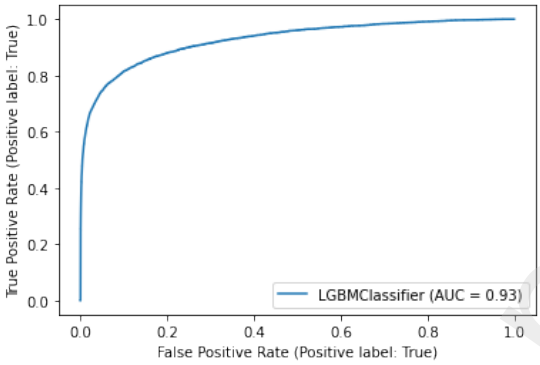
Logistic Regression



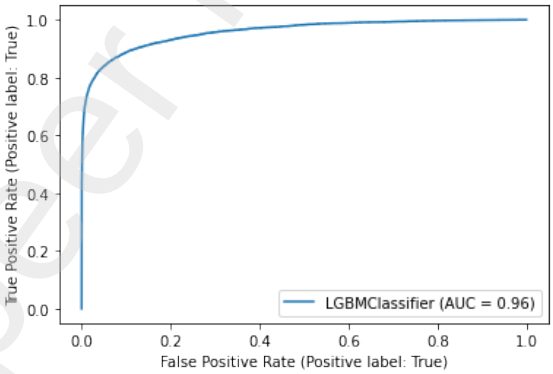
Decision Tree



Random Forest



XGBoost



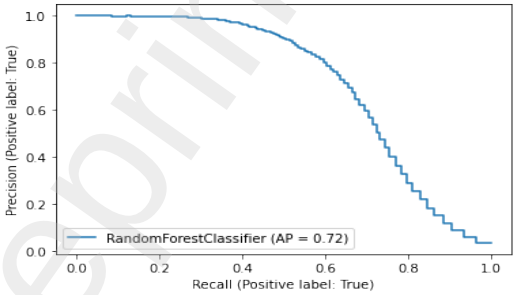
Light GBM Without Hyperparameter Tuning

Light GBM With Hyperparameter Tuning

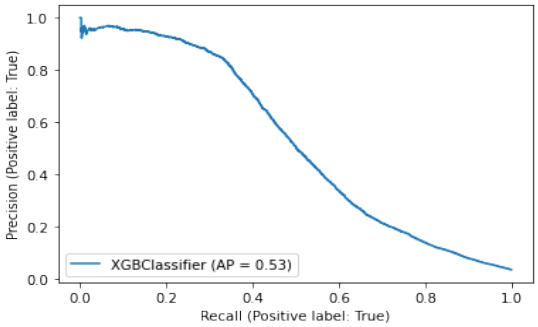
Fig. 28. AUC - ROC Curve

5.3. 3. Precision-Recall (PR) Curve

Logistic Regression



Decision Tree



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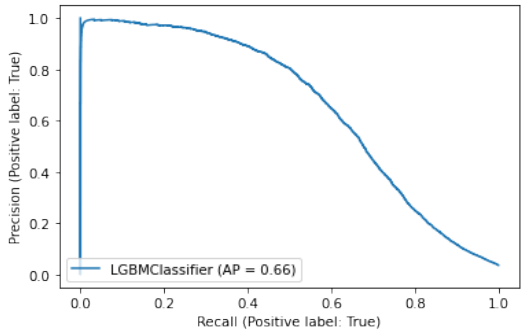
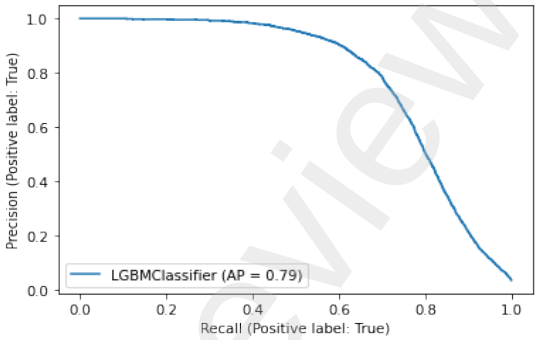
Random Forest



Light GBM Without Hyperparameter Tuning

XGBoost



Light GBM With Hyperparameter Tuning

Fig. 29. Precision-Recall (PR) Curve

5.3.4. Machine Learning Model Evaluation Summary

Table 5

Machine Learning Model Evaluation Summary

Model	Accuracy%	AUC-ROC Score	Precision	Recall	F1-Score
Logistic Regression	96.40	0.72	0.07	0.0	0.0
Decision Tree	96.39	0.77	0.50	0.56	0.53
Random Forest	97.90	0.93	0.95	0.44	0.60
XGBoost	97.31	0.88	0.88	0.29	0.43
Light GBM	97.69	0.93	0.88	0.41	0.56
Light GBM with Hyper Tuning	98.27	0.96	0.93	0.56	0.70

5.4 Deep Neural Networks Model Evaluation

This section presents the results obtained through the evaluation of implemented deep neural networks model through the figures, where Fig. 30 indicates Confusion Matrix for ANN and CNN, Fig. 31 presents Model Accuracy and Model Loss for ANN and CNN, and Table 6 presents Deep Neural Networks Model Evaluation Summary.

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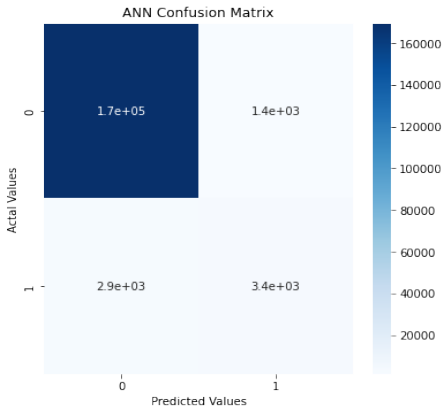
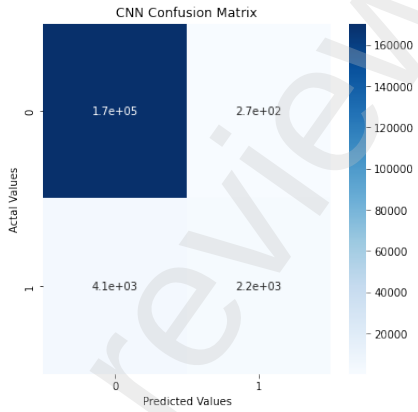
5.4.1. Confusion Matrix



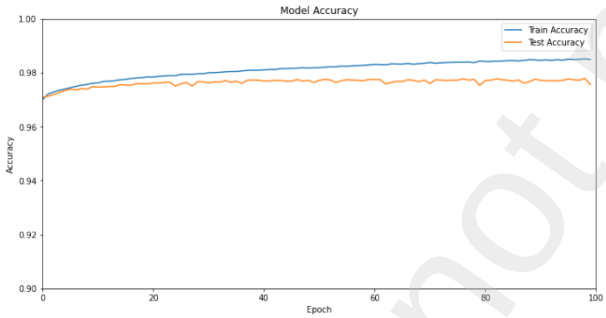
Artificial Neural Network (ANN)



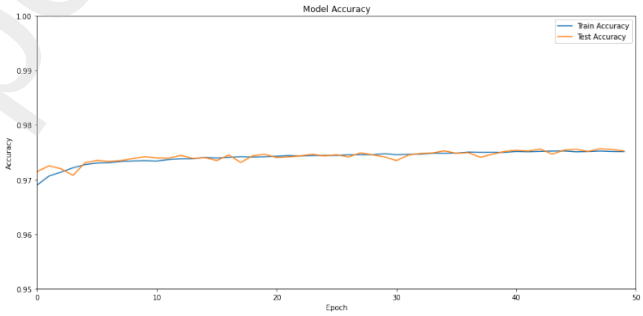
Convolutional Neural Network (CNN)

Fig. 30. Confusion Matrix for ANN and CNN

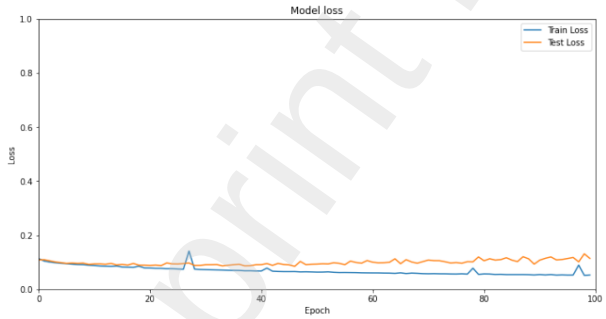
5.4.2. Model Accuracy and Model Loss



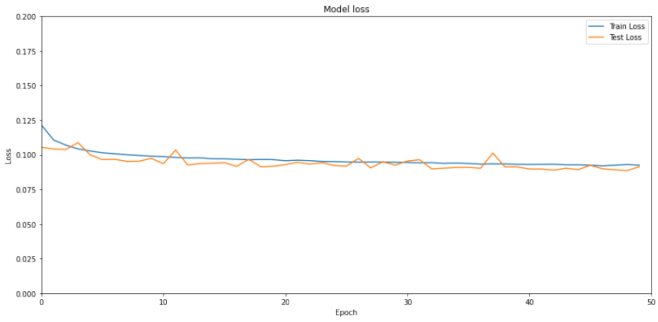
ANN – Model Accuracy



CNN – Model Accuracy



ANN – Model Loss



CNN – Model Loss

Fig. 31. Model Accuracy and Model Loss for ANN and CNN

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Table 6

Deep Neural Networks Model Evaluation Summary

Model	Accuracy%	AUC-ROC Score	Precision	Recall	F1-Score
ANN	97.56	0.76	0.71	0.54	0.61
CNN	97.52	0.67	0.89	0.35	0.50

6 Conclusion

This research tackled and filled the gap of the lack of explainability by implementing an Explainable AI (XAI)-driven Interface for Financial Transaction Fraud Detection using Machine Learning (ML) & Deep Neural Networks (DNN). Based on the literature, this is the first instance of research investigating and implementing five XAI methods like Partial Dependence Plots (PDP), Feature Importance, Shapley Additive Explanations (SHAP), SHAPASH, and Local Interpretable Model-Agnostic Explanations (LIME) on IEEE-CIS fraud detection dataset. After hyperparameter tuning, the experimental results and model evaluation with various metrics showed that the Light GBM model performed better with 98.3% Accuracy, F1- score of 0.70, and AUC-ROC score of 0.96 when compared with other ML and DNN.

The research contributed to both Explainable AI (XAI) research community and practical business value to solve fraud detection business problems in the financial services and banking industry by implementing five cutting-edge Explainable AI (XAI) methods for enhancing the model interpretability or explainability of current black-box fraud detection systems used in the Finance and Banking industry. Therefore, XAI-driven financial transaction fraud detection system decisions can be operationalized into production with greater confidence and transparency. This will save a surplus revenue, provide better forecasts and greater understandability, reduce uncertainty in business decisions, and prevent fraud in financial services and the banking industry.

New technical challenges emerge in addition to the contributions made by this research, either addressing existing shortcomings or leading to further advancements. Among them, the followings are the ones deemed to be the most logical to be explored next.

- Less Availability of Financial Data due to regulations and compliance in the industry
- Business Stakeholder Oriented Model Explainability
- Evaluation metrics for XAI Methods
- XAI Methods for Deep Neural Networks
- High computational resources like GPU and RAM are required to process large financial transaction datasets faster

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Future work can employ a cost-sensitive approach by accounting for the expenses of misclassification. False Negative costs can be several hundred to thousands of dollars higher than the penalty for designating a genuine class as fraudulent because they are significantly more expensive than the fraud amount (False Positive). This kind of learning, therefore, entails classifying an example into the category with the lowest predicted cost. The non-stationary nature of transaction distributions influences credit card theft since it forces fraudsters to devise new schemes constantly. Therefore, it is necessary to consider these changing behaviours while developing a predictive model.

Acknowledgments

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